

Chunk-wise entity lists often don't get updated correctly (Entities disappear)

MC-108469 chunk loading

Status	Resolved / Fixed
Affected	Minecraft 1.10.2, Minecraft 16w40a, Minecraft 16w43a, Minecraft 16w44a, Minecraft 1.11 Pre-Release 1, Minecraft 1.11 (+49 more)
Fixed in	20w45a
Reported	2016-10-08

STEPS TO REPRODUCE

1. Launch Minecraft and create a new superflat world.
2. In the world, execute `/setworldspawn 0 0 0``.
3. Execute `/tp 1000 100 0``.
4. Use "Esc menu -> Save and Quit to Title", then reopen the same world.
5. Execute `/summon minecraft:armor_stand 0 100 0 {NoGravity:1,Tags:["MC-108469"]}``.
6. Execute `/tp @e[tag=MC-108469,x=0,y=100,z=0,distance=..10] 200 100 0``.
7. Execute `/say @e[tag=MC-108469]`` and `/say @e[tag=MC-108469,x=200,y=100,z=0,distance=..100]``.
8. Execute `/say @e[tag=MC-108469,x=200,y=100,z=0,distance=..1000]`` and observe that this selector can find the armor stand while the previous selectors may not.
9. Use "Esc menu -> Save and Quit to Title", then reopen the same world and observe that the armor stand at approximately `200 100 0`` is gone.

OBSERVED BEHAVIOUR

Actual behavior

When an entity changes chunks, its position is updated immediately, but its membership in the per-chunk entity lists is not always updated at the same time. In affected cases, the entity remains in the list for its previous chunk until it receives another entity tick.

As a result, operations that search entities through these chunk-based lists may fail to find the entity even though:

- the entity exists;
- its coordinates have been changed to the target position;
- the target chunk is loaded; and
- the target position satisfies the selector conditions.

This is especially visible when an entity is teleported into a loaded chunk that is not currently being entity-ticked, such as a lazy chunk. The entity may not receive another tick and therefore may never be inserted into the target chunk's entity list.

For example, after teleporting the tagged armor stand from `(0, 100, 0)`` to `(200, 100, 0)`` in the reproduction steps:

```
```mcfunction
/say @e[tag=MC-108469]
...

and
```mcfunction
/say @e[tag=MC-108469,x=200,y=100,z=0,distance=..100]
...

```

may fail to find it, while:

```
```mcfunction
/say @e[tag=MC-108469,x=200,y=100,z=0,distance=..1000]

```

...

may find it. The result depends on which chunks the selector examines and on the entity-search optimizations used. It does not reflect the entity's actual coordinates.

The stale chunk association also affects saving. If the world is unloaded before the entity is added to the target chunk's list, the entity can be saved using its old chunk association. On the next reload, Minecraft reports a wrong entity location, for example:

```text

Wrong location! (12, 0) should be (0, 0), ... x=200.50, y=100.00, z=0.50

```

The entity is then not restored correctly and is lost.

The same stale-list behavior can occur when an entity is moved across multiple chunks during one tick. If a second command selects the entity using chunk-based search immediately after the first command teleports it, the second selector may not find it because the entity has not yet been registered in the relevant chunk list. This affects both command-based entity movement and movement caused by game mechanics such as pistons.

## EXPECTED BEHAVIOUR

Entities moved between chunks, whether by teleportation, pistons, or other means, should be added to the destination chunk's entity list immediately. Entity selectors and other searches should consistently find them at their actual positions, including after multiple moves in the same tick, regardless of whether the destination chunk is actively ticking. When the world is unloaded and reloaded, the entities should remain saved in and restored to the correct chunk.

## ENVIRONMENT

Minecraft Java Edition Versions: 1.10.2, 16w40a, 16w43a, 16w44a, 1.11 Pre-Release 1, 1.11, 1.11.2, 1.12, 1.12.1 Pre-Release 1, 1.12.1, 1.12.2, 17w50a, 18w20a, 18w20b, 1.13-pre1, 1.13-pre2, 1.13-pre3, 1.13-pre4, 1.13-pre6, 1.13-pre7, 1.13-pre8, 1.13, 1.13.2, 18w48a, 18w48b, 18w49a, 18w50a, 19w02a, 19w03c, 19w04a, 19w04b, 19w05a, 19w06a, 19w07a, 19w08b, 19w11b, 19w12b, 1.14.2, 1.14.3 Pre-Release 1, 1.14.3 Pre-Release 2, 1.14.3 Pre-Release 3, 1.14.3 Pre-Release 4, 1.14.3, 1.14.4 Pre-Release 1, 1.14.4 Pre-Release 2, 1.14.4 Pre-Release 3, 1.14.4 Pre-Release 4, 1.14.4, 19w36a, 19w37a, 19w40a, 19w41a, 1.15 Pre-release 1, 1.15.1, 1.15.2 Pre-Release 1 (includes snapshots)

## ORIGINAL DESCRIPTION

## The bug

In many cases when entities are moved into another chunk they are not added to the new chunks entity lists rightaway.

Instead they get added when the entity receives the next update tick. Until that happens the entity will not be found by some searches that use the chunk-wise lists.

In some cases (like in lazy chunks) it can even happen that the entity is not ticked and doesn't get added to the new list at all. Upon unloading the world the entity will not get saved in the correct chunk and be deleted upon the next reload.

Two cases where this bug can happen are teleport commands and pistons. It's likely that there are more cases though.

## How to reproduce

- Create a superflat world.

- 

```
/setworldspawn 0 0 0
```

- 

```
/tp 1000 100 0
```

- Reload the world to make sure only the right chunks are loaded.

-

```
/summon minecraft:armor_stand 0 100 0 {NoGravity:1,Tags:["MC-108469"]}
```

-

```
/tp @e[tag=MC-108469,x=0,y=100,z=0,distance=..10] 200 100 0
```

Notice that the target position is still a loaded chunk, just not one in which entities get ticked.

- If you now unload and reload the world the armour stand will be gone only leaving the following messages in the launcher:

```
[INFO] 01:58:46.893 Preparing spawn area: 0%
```

```
[WARN] 01:58:46.919 Wrong location! (12, 0) should be (0, 0), avm['Armor Stand'/22, l='19w46b', x=200.50, y=100.00, z=0.50]
```

```
[INFO] 01:58:46.960 Changing view distance to 31, from 10
```

Alternatively to reloading the world, the error can be shown using these commands. They are not able to find the entity.

```
/say @e[tag=MC-108469]
```

```
/say @e[tag=MC-108469,x=200,y=100,z=0,distance=..100]
```

In contrast this one does find the entity:

```
/say @e[tag=MC-108469,x=200,y=100,z=0,distance=..1000]
```

Given the position of the armorstand all of the above commands should be able to find it, however, since it is not constantly updated across different entity lists, only the third one will find it (as off 1.15pre1).

Also note that the behaviour of these commands is highly dependant on performance decisions of entity selectors and might not behave the same under all conditions.

This issue affects survival (pistons) as well as creative/map making (teleport commands).

## MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-108469>